

Discrete-Time Fourier Transform

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● **Discrete-Time Fourier Transform (DTFT):** Let $x(n)$ be a non-periodic signal. Its DTFT is given by

$$X(e^{j\omega}) = DTFT[x(n)]$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

The above equation usually is referred to as the synthesis equation.

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The above equation usually is referred to as the synthesis equation.

- The inverse DTFT (IDTFT) of $X(e^{j\omega})$ is given by

$$x(n) = IDTFT[X(e^{j\omega})]$$

$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

The above equation usually is referred to as the analysis equation.

Discrete-Time Fourier Transform

- The signal $x(n)$ and its DTFT $X(e^{j\omega})$ makes a Fourier transform pair. Symbolically, this is expressed as

$$x(n) \longleftrightarrow X(e^{j\omega})$$

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- In general the Fourier transform $X(e^{j\omega})$ is a complex function. It can be expressed in the polar form as

$$X(e^{j\omega}) = |X(e^{j\omega})|e^{j\angle X(e^{j\omega})}$$

- The quantity $|X(e^{j\omega})|$ is called the magnitude spectrum and it is an even function, that is,

$$|X(e^{j\omega})| = |X(e^{-j\omega})|$$

- The quantity $\angle X(e^{j\omega})$ is called the phase spectrum and it is an odd function, that is,

$$\angle X(e^{j\omega}) = -\angle X(e^{-j\omega})$$

Periodicity of DTFT

- **Periodicity of DTFT:** The DTFT $X(e^{j\omega})$ is periodic with period 2π , that is,

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$$

- *Proof:* By definition

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

For any integer k , we have

$$\begin{aligned} X(e^{j(\omega+2\pi k)}) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j(\omega+2\pi k)n} = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \underbrace{e^{-j2\pi kn}}_{=1} \\ &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \\ X(e^{j(\omega+2\pi)}) &= X(e^{j\omega}) \end{aligned}$$

Example

● **Example:** Find the Fourier transform of the causal sequence

$$x(n) = a^n u(n) \quad |a| < 1.$$

Solution: By definition

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty} a^n u(n)e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}} \end{aligned}$$

Therefore,

$$a^n u(n) \longleftrightarrow \frac{1}{1 - ae^{-j\omega}} \quad |a| < 1$$

Properties of DTFT

● **Properties of DTFT:** The pairing of $x(n)$ with its DTFT $X(e^{j\omega})$ is denoted by

$$x(n) \longleftrightarrow X(e^{j\omega})$$

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- **Linearity:** If $x_1(n) \longleftrightarrow X_1(e^{j\omega})$ and $x_2(n) \longleftrightarrow X_2(e^{j\omega})$ then

$$ax_1(n) + bx_2(n) \longleftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega})$$

- **Proof:** The Fourier transform of $ax_1(n) + bx_2(n)$ is

$$\begin{aligned}\mathcal{F}[ax_1(n) + bx_2(n)] &= \sum_{n=-\infty}^{\infty} [ax_1(n) + bx_2(n)]e^{-j\omega n} \\ &= a \underbrace{\sum_{n=-\infty}^{\infty} x_1(n)e^{-j\omega n}}_{X_1(e^{j\omega})} + b \underbrace{\sum_{n=-\infty}^{\infty} x_2(n)e^{-j\omega n}}_{X_2(e^{j\omega})} \\ &= aX_1(e^{j\omega}) + bX_2(e^{j\omega})\end{aligned}$$

Properties of DTFT contd...

Time Shifting: If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$x(n - n_o) \longleftrightarrow X(e^{j\omega})e^{-j\omega n_o}$$

Proof: The Fourier transform of $x(n - n_o)$ is

$$\mathcal{F}[x(n - n_o)] = \sum_{n=-\infty}^{\infty} x(n - n_o)e^{-j\omega n}$$

A change of variables is performed by letting $m = n - n_o$, which yields

$$\mathcal{F}[x(n - n_o)] = \sum_{m=-\infty}^{\infty} x(m)e^{-j\omega(m+n_o)} = e^{-j\omega n_o} \sum_{m=-\infty}^{\infty} x(m)e^{-j\omega m} = X(e^{j\omega})e^{-j\omega n_o}$$

Thus, the effect of a time shift on a signal is to introduce into its Fourier transform a phase shift, ωn_o , which is a linear function of ω .

Properties of DTFT contd...

● Frequency Shifting: If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$x(n)e^{j\omega_0 n} \longleftrightarrow X(e^{j(\omega-\omega_0)})$$

● **Proof:** The Fourier transform of $x(n)e^{j\omega_0 n}$ is given by

$$\begin{aligned}\mathcal{F}[x(n)e^{j\omega_0 n}] &= \sum_{n=-\infty}^{\infty} [x(n)e^{j\omega_0 n}]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} x(n)e^{-j(\omega-\omega_0)n} \\ &= X(e^{j(\omega-\omega_0)})\end{aligned}$$

Hence, a frequency shift corresponds to multiplication in time domain by a complex sinusoid whose frequency is equal to the frequency shift.

Properties of DTFT contd...

● Time Reversal: If

$$x(n) \longleftrightarrow X(e^{j\omega}),$$

then

$$y(n) = x(-n) \longleftrightarrow Y(e^{j\omega}) = X(e^{-j\omega})$$

● Proof: By definition

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x(-n)e^{-j\omega n}$$

Substituting $m = -n$ into the above equation, we obtain

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(m)e^{j\omega m} \\ &= \sum_{n=-\infty}^{\infty} x(m)e^{-j(-\omega)m} = X(e^{-j\omega}) \end{aligned}$$

Properties of DTFT contd...

- **Case-I:** $x(n)$ is even. If $x(n)$ is even then its Fourier transform is also even.

$$\begin{aligned} \text{if } x(n) &= x(-n) \\ \text{then } X(e^{j\omega}) &= X(e^{-j\omega}) \end{aligned}$$

- **Case-II:** $x(n)$ is odd. If $x(n)$ is odd then its Fourier transform is also odd.

$$\begin{aligned} \text{if } x(n) &= -x(-n) \\ \text{then } X(e^{j\omega}) &= -X(e^{-j\omega}) \end{aligned}$$

- **Time Expansion:** Let m be a positive integer, and define the signal

$$x_{(m)}(n) = \begin{cases} x\left(\frac{n}{m}\right) & \text{if } n \text{ is a multiple of } m \\ 0 & \text{if } n \text{ is not a multiple of } m. \end{cases}$$

$x_{(m)}(n)$ can be obtained from $x(n)$ by placing $m - 1$ zeros between successive values of the original signal.

Properties of DTFT contd...

Now, If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then,

$$y(n) = x_{(m)}(n) \longleftrightarrow Y(e^{j\omega}) = X(e^{jm\omega})$$

Proof: By definition

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_{(m)}(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x\left(\frac{n}{m}\right)e^{-j\omega n}$$

A change of variables is performed by letting $r = \frac{n}{m}$, which also yields $r = -\infty$ as $n = -\infty$, and $r = \infty$ as $n = \infty$. Therefore,

$$Y(e^{j\omega}) = \sum_{r=-\infty}^{\infty} x(r)e^{-j\omega mr} = X(e^{jm\omega})$$

Properties of DTFT contd...

● **First Difference:** The first-difference operation is defined as $y(n) = x(n) - x(n - 1)$. Now, if

$$x(n) \longleftrightarrow X(e^{j\omega}),$$

then

$$y(n) = x(n) - x(n - 1) \longleftrightarrow Y(e^{j\omega}) = (1 - e^{-j\omega})X(e^{j\omega})$$

● **Proof:** Using the time-shifting property, we get

$$\text{if } x(n) \longleftrightarrow X(e^{j\omega}) \quad \text{then } x(n - 1) \longleftrightarrow X(e^{j\omega})e^{-j\omega}$$

Now, using the linearity property, we get

$$x(n) - x(n - 1) \longleftrightarrow X(e^{j\omega}) - X(e^{j\omega})e^{-j\omega}$$

$$x(n) - x(n - 1) \longleftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$$

Properties of DTFT contd...

● Differentiation in Frequency Domain: If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$-jnx(n) \longleftrightarrow \frac{dX(e^{j\omega})}{d\omega}$$

or

$$nx(n) \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

● **Proof:** By definition

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

Differentiating both sides w.r.t. ω , we obtain

$$\frac{dX(e^{j\omega})}{d\omega} = \sum_{n=-\infty}^{\infty} [-jnx(n)]e^{-j\omega n}$$

Properties of DTFT contd...

$$\frac{dX(e^{j\omega})}{d\omega} = \mathcal{F}[-jnx(n)]$$

Therefore,

$$-jnx(n) \longleftrightarrow \frac{dX(e^{j\omega})}{d\omega}$$

Multiplying both sides by j , we obtain

$$nx(n) \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

Properties of DTFT contd...

● Convolution Property: If

$$x_1(n) \longleftrightarrow X_1(e^{j\omega}) \quad \text{and} \quad x_2(n) \longleftrightarrow X_2(e^{j\omega})$$

then

$$x_1(n) * x_2(n) \longleftrightarrow X_1(e^{j\omega})X_2(e^{j\omega})$$

The Fourier transform maps the convolution of two signals into the product of their Fourier transforms.

● **Proof:** The Fourier transform of $x_1(n) * x_2(n)$ is

$$\begin{aligned} \mathcal{F}[x_1(n) * x_2(n)] &= \sum_{n=-\infty}^{\infty} [x_1(n) * x_2(n)] e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \left(\sum_{m=-\infty}^{\infty} x_1(m) x_2(n-m) \right) e^{-j\omega n} \end{aligned}$$

Properties of DTFT contd...

By interchanging the order of summation, we have

$$\mathcal{F}[x_1(n) * x_2(n)] = \sum_{m=-\infty}^{\infty} x_1(m) \underbrace{\left(\sum_{n=-\infty}^{\infty} x_2(n-m)e^{-j\omega n} \right)}_{=X_2(e^{j\omega})e^{-j\omega m}}$$

$$\mathcal{F}[x_1(n) * x_2(n)] = \sum_{m=-\infty}^{\infty} x_1(m) \left(X_2(e^{j\omega})e^{-j\omega m} \right)$$

$$= X_2(e^{j\omega}) \sum_{m=-\infty}^{\infty} x_1(m)e^{-j\omega m}$$

$$\mathcal{F}[x_1(n) * x_2(n)] = X_2(e^{j\omega})X_1(e^{j\omega}) = X_1(e^{j\omega})X_2(e^{j\omega})$$

Therefore,

$$x_1(n) * x_2(n) \longleftrightarrow X_1(e^{j\omega})X_2(e^{j\omega})$$

Properties of DTFT contd...

● Accumulation Property: If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$\sum_{k=-\infty}^n x(k) \longleftrightarrow \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m)$$

● **Proof:** Convolution a signal $x(n)$ with a unit step function $u(n)$, we obtain

$$x(n) * u(n) = \sum_{k=-\infty}^{\infty} x(k)u(n-k)$$

Since

$$u(n-k) = \begin{cases} 1 & n-k \geq 0 \longrightarrow k \leq n \\ 0 & n-k < 0 \longrightarrow k > n \end{cases}$$

Properties of DTFT contd...

we can write

$$x(n) * u(n) = \sum_{k=-\infty}^n x(k)$$

Now we can prove the accumulation property of the Fourier transform.

$$\sum_{k=-\infty}^n x(k) = x(n) * u(n)$$
$$\mathcal{F} \left[\sum_{k=-\infty}^n x(k) \right] = \mathcal{F}[x(n) * u(n)]$$

Using the convolution property, we obtain

$$\mathcal{F} \left[\sum_{k=-\infty}^n x(k) \right] = X(e^{j\omega})U(e^{j\omega}) = X(e^{j\omega}) \left(\frac{1}{1 - e^{-j\omega}} + \pi \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m) \right)$$

Properties of DTFT contd...

$$\mathcal{F} \left[\sum_{k=-\infty}^n x(k) \right] = X(e^{j\omega})U(e^{j\omega}) = \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j\omega}) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m)$$
$$\mathcal{F} \left[\sum_{k=-\infty}^n x(k) \right] = \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m)$$

Therefore,

$$\sum_{k=-\infty}^n x(k) \longleftrightarrow \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m)$$

Properties of DTFT contd...

● Multiplication (or Modulation) Property: If

$$x_1(n) \longleftrightarrow X_1(e^{j\omega}) \quad \text{and} \quad x_2(n) \longleftrightarrow X_2(e^{j\omega})$$

then

$$x_1(n)x_2(n) \longleftrightarrow \frac{1}{2\pi} [X_1(e^{j\omega}) \circledast X_2(e^{j\omega})] = \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) X_2(e^{j(\omega-\theta)}) d\theta$$

where $\circledast$ denotes the periodic convolution.

● **Proof:** The Fourier transform of $x_1(n)x_2(n)$ is given by

$$\begin{aligned} \mathcal{F}[x_1(n)x_2(n)] &= \sum_{n=-\infty}^{\infty} [x_1(n)x_2(n)]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta})e^{j\theta n} d\theta \right) x_2(n)e^{-j\omega n} \end{aligned}$$

Properties of DTFT contd...

$$\mathcal{F}[x_1(n)x_2(n)] = \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) \underbrace{\left(\sum_{n=-\infty}^{\infty} [x_2(n)e^{j\theta n}] e^{-j\omega n} \right)}_{X_2(e^{j(\omega-\theta)})} d\theta$$
$$\mathcal{F}[x_1(n)x_2(n)] = \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) X_2(e^{j(\omega-\theta)}) d\theta = \frac{1}{2\pi} [X_1(e^{j\omega}) \circledast X_2(e^{j\omega})]$$

Therefore,

$$x_1(n)x_2(n) \longleftrightarrow \frac{1}{2\pi} [X_1(e^{j\omega}) \circledast X_2(e^{j\omega})]$$

The Fourier transform maps the multiplication of two signals into the convolution of their Fourier transforms.

Properties of DTFT contd...

● Conjugation and Conjugate Symmetry: If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$x^*(n) \longleftrightarrow X^*(e^{-j\omega})$$

● **Proof:** By definition

$$\begin{aligned}\mathcal{F}[x^*(n)] &= \sum_{n=-\infty}^{\infty} x^*(n) e^{-j\omega n} \\ &= \left[\sum_{n=-\infty}^{\infty} x(n) e^{j\omega n} \right]^* \\ &= \left[\sum_{n=-\infty}^{\infty} x(n) e^{-j(-\omega)n} \right]^* \\ \mathcal{F}[x^*(n)] &= [X(e^{-j\omega})]^* = X^*(e^{-j\omega})\end{aligned}$$

Properties of DTFT contd...

🟢 **Case I:** If $x(n)$ is real, that is, if

$$x^*(n) = x(n)$$

then $\mathcal{F}[x^*(n)] = \mathcal{F}[x(n)]$

$$X^*(e^{-j\omega}) = X(e^{j\omega})$$

$$X(e^{-j\omega}) = X^*(e^{j\omega}) \quad (\text{conjugate symmetric})$$

The DTFT of a real signal is *conjugate symmetric*.

🟢 **Case II:** If $x(n)$ is *real & even*, that is, if

$$x(n) = x^*(n) = x(-n)$$

then $\mathcal{F}[x(n)] = \mathcal{F}[x^*(n)] = \mathcal{F}[x(-n)]$

$$X(e^{j\omega}) = X^*(e^{-j\omega}) = X(e^{-j\omega}) \quad (\text{real & even})$$

The DTFT of a real & even signal is real & even.

Properties of DTFT contd...

● **Case III:** If $x(n)$ is *real & odd*, that is, if

$$x(n) = x^*(n) = -x(-n)$$

then $\mathcal{F}[x(n)] = \mathcal{F}[x^*(n)] = \mathcal{F}[-x(-n)]$

$$X(e^{j\omega}) = X^*(e^{-j\omega}) = -X(e^{-j\omega}) \quad (\text{imaginary \& odd})$$

The DTFT of a real & odd signal is purely imaginary & odd.

● **Case IV:** If $x(n)$ is *imaginary*, that is, if

$$x^*(n) = -x(n)$$

then $\mathcal{F}[x^*(n)] = \mathcal{F}[-x(n)]$

$$X^*(e^{-j\omega}) = -X(e^{j\omega})$$

$$X(e^{-j\omega}) = -X^*(e^{j\omega}) \quad (\text{conjugate antisymmetric})$$

The DTFT of an *imaginary* signal is *conjugate antisymmetric*.

Properties of DTFT contd...

● **Case V:** If $x(n)$ is *imaginary & even*, that is, if

$$x(n) = -x^*(n) = x(-n)$$

then $\mathcal{F}[x(n)] = \mathcal{F}[-x^*(n)] = \mathcal{F}[x(-n)]$

$$X(e^{j\omega}) = -X^*(e^{-j\omega}) = X(e^{-j\omega}) \quad (\text{imaginary \& even})$$

The DTFT of an imaginary & even signal is imaginary & even.

● **Case VI:** If $x(n)$ is *imaginary & odd*, that is, if

$$x(n) = -x^*(n) = -x(-n)$$

then $\mathcal{F}[x(n)] = \mathcal{F}[-x^*(n)] = \mathcal{F}[-x(-n)]$

$$X(e^{j\omega}) = -X^*(e^{-j\omega}) = -X(e^{-j\omega}) \quad (\text{imaginary \& odd})$$

The DTFT of an imaginary & odd signal is real & odd.

Properties of DTFT contd...

- **Case VII:** Even and odd decomposition of real signals: Let $x(n)$ be a real signal and

$$x(n) \longleftrightarrow X(e^{j\omega})$$

- The DTFT of the even part of $x(n)$ is the real value of $X(e^{j\omega})$, that is,

$$x_e(n) = \mathcal{E}\{x(n)\} \longleftrightarrow \text{Re}\{X(e^{j\omega})\} = X_R(e^{j\omega})$$

The even part of a signal $x(n)$ is defined as

$$x_e(n) = \frac{1}{2}[x(n) + x(-n)]$$

$$\mathcal{F}[x_e(n)] = \mathcal{F}\left[\frac{1}{2}[x(n) + x(-n)]\right] = \frac{1}{2}\left(\mathcal{F}[x(n)] + \mathcal{F}[x(-n)]\right) = \frac{1}{2}\left(X(e^{j\omega}) + X^*(e^{j\omega})\right)$$

$$\mathcal{F}[x_e(n)] = \frac{1}{2}2X_R(e^{j\omega}) = X_R(e^{j\omega})$$

Therefore, $x_e(n) = \mathcal{E}\{x(n)\} \longleftrightarrow X_R(e^{j\omega})$

Properties of DTFT contd...

- 🟡 The DTFT of the odd part of $x(n)$ is $jX_I(e^{j\omega})$, that is,

$$x_o(n) = \mathcal{O}\{x(n)\} \longleftrightarrow j\text{Im}\{X(e^{j\omega})\} = jX_I(e^{j\omega})$$

The odd part of a signal $x(n)$ is defined as

$$x_o(n) = \frac{1}{2}[x(n) - x(-n)]$$

$$\mathcal{F}[x_o(n)] = \mathcal{F}\left[\frac{1}{2}[x(n) - x(-n)]\right] = \frac{1}{2}\left(\mathcal{F}[x(n)] - \mathcal{F}[x(-n)]\right)$$

$$= \frac{1}{2}\left(X(e^{j\omega}) - X^*(e^{j\omega})\right)$$

$$\mathcal{F}[x_o(n)] = \frac{1}{2}2jX_I(e^{j\omega}) = jX_I(e^{j\omega})$$

Therefore,

$$x_o(n) = \mathcal{O}\{x(n)\} \longleftrightarrow jX_I(e^{j\omega})$$

Properties of DTFT contd...

● **Parseval's Relation:** Let $x(n)$ be an energy signal and if

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

● **Proof:** Consider the LHS, we have

$$\begin{aligned} E_x &= \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=-\infty}^{\infty} x(n)x^*(n) = \sum_{n=-\infty}^{\infty} x(n) \left(\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \right)^* \\ &= \sum_{n=-\infty}^{\infty} x(n) \left(\frac{1}{2\pi} \int_{2\pi} X^*(e^{j\omega}) e^{-j\omega n} d\omega \right) = \frac{1}{2\pi} \int_{2\pi} X^*(e^{j\omega}) \left(\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right) d\omega \\ &= \frac{1}{2\pi} \int_{2\pi} X^*(e^{j\omega}) X(e^{j\omega}) d\omega = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega \end{aligned}$$

Fourier Transform of Periodic Signals

- **Fourier Transform of Periodic Signals:** Consider the periodic signal $x(n)$ with period N and fundamental frequency $\omega_0 = \frac{2\pi}{N}$. Its DTFT is given by

$$X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} X_k \delta(\omega - k\omega_0)$$

- **Proof:** The signal $x(n)$ has the discrete-time Fourier series representation

$$x(n) = \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n}$$

Taking the Fourier transform on both sides, we get

$$\begin{aligned} \mathcal{F}[x(n)] &= \mathcal{F} \left[\sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \right] \\ X(e^{j\omega}) &= \sum_{k=\langle N \rangle} X_k \mathcal{F}[e^{jk\omega_0 n}] \end{aligned}$$

Fourier Transform of Periodic Signals contd...

Since

$$\mathcal{F}[e^{jk\omega_0 n}] = 2\pi\delta(\omega - k\omega_0), \quad 0 \leq \omega < 2\pi$$

we get

$$X(e^{j\omega}) = \sum_{k=\langle N \rangle} X_k 2\pi\delta(\omega - k\omega_0), \quad 0 \leq \omega < 2\pi$$

$$X(e^{j\omega}) = 2\pi \sum_{k=\langle N \rangle} X_k \delta(\omega - k\omega_0), \quad 0 \leq \omega < 2\pi$$

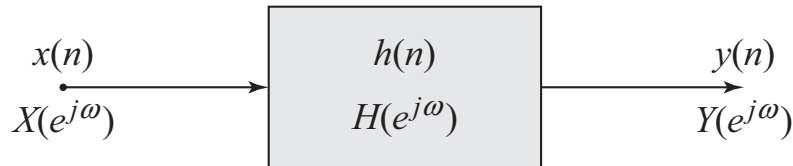
Since the DTFT is periodic with period 2π . It follows that $X(e^{j\omega})$ consists a set of N impulses of strength $2\pi X_k$, $k = 0, 1, 2, \dots, N - 1$, repeated at intervals of $N\omega_0 = 2\pi$. Thus, $X(e^{j\omega})$ can be compactly written as

$$X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} X_k \delta(\omega - k\omega_0)$$

Signal Transmission Through LTI Systems

- **Signal Transmission Through LTI Systems:** The output of an LTI system is given by

$$y(n) = x(n) * h(n)$$



Using DTFT, we can write

$$\mathcal{F}[y(n)] = \mathcal{F}[x(n)] * \mathcal{F}[h(n)]$$

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

The frequency response of an LTI system is given by

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

Signal Transmission Through LTI Systems

- Expressing the above equation in polar form, we get

$$|Y(e^{j\omega})|e^{j\angle Y(e^{j\omega})} = [|X(e^{j\omega})|e^{j\angle X(e^{j\omega})}] [|H(e^{j\omega})|e^{j\angle H(e^{j\omega})}]$$
$$|Y(e^{j\omega})|e^{j\angle Y(e^{j\omega})} = |X(e^{j\omega})||H(e^{j\omega})|e^{j[\angle X(e^{j\omega}) + \angle H(e^{j\omega})]}$$

Therefore,

$$|Y(e^{j\omega})| = |X(e^{j\omega})||H(e^{j\omega})|$$

and

$$\angle Y(e^{j\omega}) = \angle X(e^{j\omega}) + \angle H(e^{j\omega})$$

- During transmission, the input signal magnitude spectrum $|X(e^{j\omega})|$ is changed to $|X(e^{j\omega})||H(e^{j\omega})|$.
- The input signal phase spectrum $\angle X(e^{j\omega})$ is changed to $[\angle X(e^{j\omega}) + \angle H(e^{j\omega})]$.
- An input signal spectral component of frequency ω is modified in amplitude by a factor $|H(e^{j\omega})|$ and is shifted in phase by an angle $\angle H(e^{j\omega})$.
- $|H(e^{j\omega})|$ is the magnitude response.
- $\angle H(e^{j\omega})$ is the phase response of the system.

Response to Sinusoidal Signals

● **Response to Sinusoidal Signals:** Let the input to the LTI system be

$$x(n) = A \cos(\omega_0 n), \quad -\infty < n < \infty$$

$$x(n) = \frac{A}{2} e^{j\omega_0 n} + \frac{A}{2} e^{-j\omega_0 n}$$

Because of the linearity, the response $y(n)$ to the input $x(n)$ is given by

$$\begin{aligned} y(n) &= \frac{A}{2} H(e^{j\omega_0}) e^{j\omega_0 n} + \frac{A}{2} H(e^{-j\omega_0}) e^{-j\omega_0 n} \\ &= \frac{A}{2} |H(e^{j\omega_0})| e^{j\angle H(e^{j\omega_0})} e^{j\omega_0 n} + \frac{A}{2} |H(e^{-j\omega_0})| e^{j\angle H(e^{-j\omega_0})} e^{-j\omega_0 n} \end{aligned}$$

Since $|H(e^{j\omega_0})| = |H(e^{-j\omega_0})|$, and $\angle H(e^{j\omega_0}) = -\angle H(e^{-j\omega_0})$, we obtain

$$\begin{aligned} y(n) &= \frac{A}{2} |H(e^{j\omega_0})| e^{j\angle H(e^{j\omega_0})} e^{j\omega_0 n} + \frac{A}{2} |H(e^{j\omega_0})| e^{-j\angle H(e^{j\omega_0})} e^{-j\omega_0 n} \\ y(n) &= \frac{A}{2} |H(e^{j\omega_0})| e^{j(\omega_0 n + \angle H(e^{j\omega_0}))} + \frac{A}{2} |H(e^{j\omega_0})| e^{-j(\omega_0 n + \angle H(e^{j\omega_0}))} \end{aligned}$$

Response to Sinusoidal Signals contd...



$$y(n) = A |H(e^{j\omega_0})| \cos[\omega_0 n + \angle H(e^{j\omega_0})] = A |H(e^{j\omega_0})| \cos[\omega_0 n + \theta(\omega_0)]$$



Thus, the output signal $y(n)$ has the same sinusoidal waveform as the input signal $x(n)$, with two differences:



(1) The amplitude is multiplied by $|H(e^{j\omega_0})|$, the value of the magnitude response of the discrete-time system at $\omega = \omega_0$.

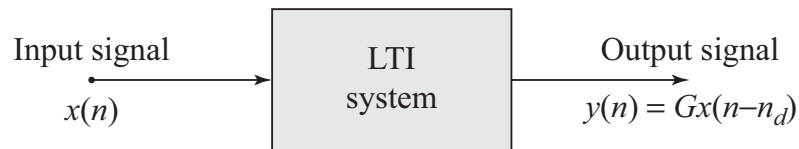


(2) The output signal has a phase lag relative to the input by an amount $\theta(\omega_0) = \angle H(e^{j\omega_0})$, the value of the phase response of the discrete-time system at $\omega = \omega_0$.

$$A \cos(\omega_0 n) \longrightarrow \boxed{H(e^{j\omega})} \longrightarrow A |H(e^{j\omega_0})| \cos[\omega_0 n + \theta(\omega_0)]$$

Linear and Nonlinear Phase

- **Linear and Nonlinear Phase:** In several applications, such as signal amplification or message signal transmission over a communication channel, we require that the output waveform be a replica of the input waveform, i.e., distortionless transmission.



- Transmission is said to be distortionless if the input and the output have identical wave-shapes within a multiplicative constant and a constant time delay.
- Thus, in distortionless transmission, the input $x(n)$ and the output $y(n)$ satisfy the condition

$$y(n) = Gx(n - n_d)$$

- The constant G accounts for a change in amplitude and the constant n_d accounts for a delay in transmission.

Linear and Nonlinear Phase contd...

- Consider the equation.

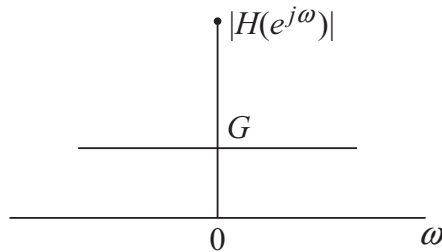
$$y(n) = Gx(n - n_d)$$

- Taking the Fourier transform of the above equation, we get

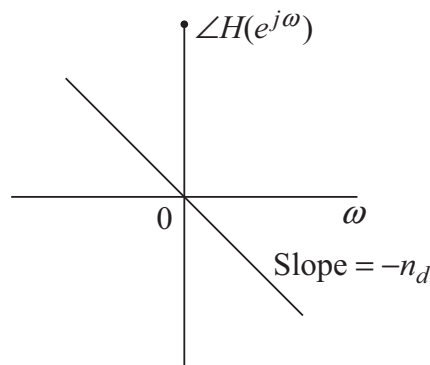
$$Y(e^{j\omega}) = GX(e^{j\omega})e^{-j\omega n_d}$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = H(e^{j\omega}) = Ge^{-j\omega n_d}$$

This is the frequency response required of a system for distortionless transmission.



(a)



(b)

Linear and Nonlinear Phase contd...

- The magnitude response $|H(e^{j\omega})|$ must be a constant; that is, we must have

$$|H(e^{j\omega})| = G$$

for some constant G .

- The phase response $\angle H(e^{j\omega})$ must be a linear function of ω with slope $-n_d$ and intercept zero; that is, we must have

$$\angle H(e^{j\omega}) = -\omega n_d$$

- A linear phase $\angle H(e^{j\omega}) = -\omega n_d$ means that every spectral component is delayed by n_d time units.
- For distortionless transmission, we require a linear phase characteristic.

Phase Delay and Group Delay

● **Phase Delay and Group Delay:** For a linear phase system (whose phase varies linearly with frequency), both the phase delay and the group delay are constant. These two parameters are associated with the frequency response $H(e^{j\omega})$ of the system.

● **Phase Delay:** The time delay experienced by a single frequency signal (that is, a sinusoidal signal) when the signal passes through a system is referred to as the system phase delay and is defined as

$$\tau_p(\omega) = -\frac{\angle H(e^{j\omega})}{\omega} = -\frac{\theta(\omega)}{\omega}$$

Let the input signal to an LTI system be $x(n) = A \cos(\omega_0 n + \theta)$

Then the output is $y(n) = A |H(e^{j\omega_0})| \cos(\omega_0(n - \tau_p(\omega_0)) + \theta)$

where

$$\tau_p(\omega) \Big|_{\omega=\omega_0} = \tau_p(\omega_0) = -\frac{\angle H(e^{j\omega_0})}{\omega_0} = -\frac{\theta(\omega_0)}{\omega_0}$$

Phase Delay and Group Delay contd...

- **Group Delay:** The group delay is defined as

$$\tau_g(\omega) = -\frac{d\angle H(e^{j\omega})}{d\omega} = -\frac{d\theta(\omega)}{d\omega}$$

- The time delay $\tau_g(\omega)$ is called the group delay or envelope delay. Thus, the group delay at each frequency equals the negative of the slope of the phase at that frequency.
- If $\tau_g(\omega)$ is constant, all the components are delayed by the same interval.
- When the input signal contains many sinusoidal components with different frequencies that are not harmonically related, each component will go through different phase delays when processed by a frequency selective LTI system, and the signal delay is determined using group delay.
- Let the input signal to an LTI system be

$$s(n) = A \cos(\omega_m n) \cos(\omega_c n)$$

Phase Delay and Group Delay contd...

The system output signal is given by

$$y(n) = A \cos\left(\omega_c [n - \tau_p(\omega_c)]\right) \cos\left(\omega_m [n - \tau_g(\omega_c)]\right)$$

where

$$\tau_p(\omega_c) = -\frac{\theta(\omega_c)}{\omega_c}$$

and

$$\tau_g(\omega_c) = -\left.\frac{d\theta(\omega)}{d\omega}\right|_{\omega=\omega_c} = -\left.\frac{d\angle H(\omega)}{d\omega}\right|_{\omega=\omega_c}$$

DTFT

Thank You!